



Cloud Battle

Author : Redhouane Abdellah

Difficulty Level : Medium

Cloud Battle

Raouf is on a national mission to the capital of Uzbekistan, Tashkent. During his journey, he gets bored and starts watching n clouds aligned in the sky, numbered from 1 to n . To pass the time, he begins a game involving a “cloudy score” based on the segments of clouds he observes.

He uses a recursive procedure to look for the cloud in the middle of each segment $[l, r]$, but his responsibility score k prevents him from observing any segment with a length strictly less than k . Not wanting to play alone, he challenges Hachem, who follows the same rules with N clouds and a responsibility score K .

Task: Determine the winner of the game based on who achieves the higher cloudy score, or determine if the game ends in a tie.

Rules of the Game

For a segment $[l, r]$ with length $len = r - l + 1$:

- If $len < k$, the observation stops for that segment.
- Calculate the middle index $m = \lfloor \frac{l+r}{2} \rfloor$.
- If the length of the segment is **even**, the segment is divided into two segments of the same length, $[l, m]$ and $[m + 1, r]$, for further observation. No points are added to the score.
- If the length of the segment is **odd**, the cloudy score increases by m . If $l \neq r$, the observation continues with the two segments $[l, m - 1]$ and $[m + 1, r]$.
- He starts with the segment $[1, n]$ and continues to observe segments adhering to the other rules.

Input

The first line of input contains two integers n and k ($1 \leq k \leq n \leq 2 \cdot 10^9$), representing the number of clouds Raouf sees and his responsibility score, respectively.

The second line of input contains two integers N and K ($1 \leq K \leq N \leq 2 \cdot 10^9$), representing the number of clouds Hachem sees and his responsibility score, respectively.

Output

If Raouf has a bigger cloudy score than Hachem, print “Raouf”.

If Hachem has a bigger cloudy score than Raouf, print “Hachem”.

Otherwise, print "Tie".

Constraints

- $1 \leq k \leq n \leq 2 \cdot 10^9$
- $1 \leq K \leq N \leq 2 \cdot 10^9$
- Time limit: **1.0 seconds**
- Memory limit: **256 MB**

Sample Input 1

```
5 1
5 2
```

Sample Output 1

Raouf

Explanation

For Raouf ($n = 5, k = 1$):

- Segment $[1, 5]$ has odd length (5). $m = 3$. Score becomes 3.
- Next segments are $[1, 2]$ and $[4, 5]$. Both have even length (2), so they split without adding score.
- Since $k = 1$, Raouf processes the resulting segments $[1, 1], [2, 2], [4, 4], [5, 5]$, adding $1 + 2 + 4 + 5 = 12$ to the score.
- **Total score for Raouf:** $3 + 12 = 15$.

For Hachem ($N = 5, K = 2$):

- Segment $[1, 5]$ has odd length (5). $m = 3$. Score becomes 3.
- Next segments $[1, 2]$ and $[4, 5]$ have length 2. Since $K = 2$, they are processed (even length, no score) and split into segments of length 1.
- However, segments of length 1 are not observed because $1 < K$.
- **Total score for Hachem:** 3.

Since $15 > 3$, Raouf wins.

Sample Input 2

```
3 2
3 3
```

Sample Output 2

Tie